



CONTINGENCY LESSONS

Mathematics for SylvanSync™

Advanced



GRADES 9-12

TO BE USED ONLY IN CASE OF EMERGENCY, WITH THE APPROVAL OF TECHNOLOGY SERVICES.

Mathematics for SylvanSync™

Grades 9–12

VersaTiles: Integer Operations

Objectives

The student will be able to:

- add and subtract integers.
- multiply and divide integers.
- solve problems involving integers.

Materials

- *VersaTiles: Integer Operations* book (VT7IO)
- VersaTiles

Background Information

Example:

At 7:00 p.m. the temperature was -3°C . By 1:00 a.m. the temperature had dropped to -12°C . How many degrees Celsius did the temperature change from 7:00 p.m. to 1:00 a.m.?

To determine how many degrees the temperature changed, the difference in the temperatures must be found. Recall that subtracting a negative can be rewritten as addition:

$$-12 - (-3) = -12 + 3 = -9$$

So, the change in temperature was -9°C . A change in temperature of -9°C can also be thought of as a decrease of 9°C since it is negative.

If the temperature decreased 9°C from 7:00 p.m. to 1:00 a.m. then, assuming a constant decrease, how many degrees Celsius did it decrease each hour?

In order to answer this question, first determine the number of hours from 7:00 p.m. to 1:00 a.m., which is six hours. So, the temperature decreased 9°C over the course of six hours which is equivalent to:

$$\frac{-9^{\circ}\text{C}}{6 \text{ hours}}$$

Divide -9 by 6 to determine the decrease per hour.

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VersaTiles: Integer Operations

An **integer** is the set of positive whole numbers, their opposites, and zero. Remember the rules for multiplying and dividing integers:

- When multiplying or dividing two integers with like signs the result is positive.
- When multiplying or dividing two integers with unlike signs the result is negative.

$$\frac{-9^{\circ}\text{C}}{6 \text{ hours}} = \frac{-1.5^{\circ}\text{C}}{1 \text{ hour}}$$

The temperature decreased 1.5°C each hour or changed -1.5°C per hour.

Activity

Go to page 8 in *VersaTiles: Integer Operations*. Match the addition problem to the sum. Flip over the box to see if you did it correctly.

Go to page 15 in *VersaTiles: Integer Operations*. Match the subtraction problem to the answer. Flip over the box to see if you did it correctly.

Go to page 23 in *VersaTiles: Integer Operations*. Match the multiplication problem to the product. Flip over the box to see if you did it correctly.

Go to page 26 in *VersaTiles: Integer Operations*. Match the division problem to the quotient. Flip over the box to see if you did it correctly.

Go to pages 18–19 in *VersaTiles: Integer Operations*. Match each problem to the answer. Flip over the box to see if you did it correctly.

Go to pages 24–25 in *VersaTiles: Integer Operations*. Match each problem to the answer. Flip over the box to see if you did it correctly.

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Grades 9–12

VersaTiles: Algebra and Functions

Objectives

The student will be able to:

- write an equation to represent a situation.
- solve one-step equations.
- solve two-step equations.

Materials

- *VersaTiles: Algebra and Functions* book (VT8AF)
- VersaTiles

Background Information

Example 1:

Michaela is saving money to buy a new bicycle. The bicycle costs \$100. If Michaela saves \$15 each week, after how many weeks will she have enough money to purchase the bicycle?

The unknown in the problem is the number of weeks. A **variable** is a letter used to represent an unknown quantity. Let the variable w represent the number of weeks Michaela saves money. To help write an equation for the situation, take a look at the table.

Number of Weeks	Money Saved
1	\$15
2	\$30
3	\$45
w	$\$(15w)$

After w weeks, Michaela has $\$(15w)$. To determine when she will have enough money to buy the bicycle the equation $15w = 100$ can be used.

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Grades 9–12
VersaTiles: Algebra and Functions

Inverse operations are two operations that “undo” each other. One pair is multiplication and division. The other pair of inverse operations is addition and subtraction. To solve the equation $15w = 100$, perform inverse operations on both sides of the equation to isolate the variable w .

$$15w = 100$$

$$\frac{15w}{15} = \frac{100}{15}$$

$$w \approx 6.7$$

After six weeks, Michaela will have $6(\$15)$ or \$90, so the answer 6.7 should be rounded up to seven. Therefore, after seven weeks of saving \$15, Michaela will have enough money to purchase the bicycle.

Example 2:

Now suppose Michaela’s parents say they will give her \$20 towards the purchase of the bicycle. How many weeks will it take Michaela to save the money for the bicycle?

Since Michaela will have \$20 from her parents, the equation $15w + 20 = 100$ can be used to solve the problem.

The first step in solving for w is to subtract 20 from both sides of the equation.

$$15w + 20 - 20 = 100 - 20$$

$$15w = 80$$

Finally, divide both sides of the equation by 15.

$$\frac{15w}{15} = \frac{80}{15}$$

$$w \approx 5.3$$

So, if her parents give her \$20 towards the bicycle purchase, it will take her 6 weeks to have enough money.

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VersaTiles: Algebra and Functions

Example 3:

Solve for x :

$$\frac{x}{-3} - 7 = -11$$

Since seven is being subtracted from the variable term, add seven to both sides of the equation.

$$\frac{x}{-3} - 7 + 7 = -11 + 7$$

$$\frac{x}{-3} = -4$$

To isolate x , multiply both sides of the equation by -3 because x is being divided by -3 .

$$3 \cdot \frac{x}{-3} = -4 \cdot 3$$

$$x = -12$$

Activity

Go to page 7 in *VersaTiles: Algebra and Functions*. Match each equation to the solution. Flip over the box to see if you did it correctly.

Go to page 9 in *VersaTiles: Algebra and Functions*. Match each equation to the solution. Flip over the box to see if you did it correctly.

Go to page 16 in *VersaTiles: Algebra and Functions*. Match each equation to the solution. Flip over the box to see if you did it correctly.

Go to page 17 in *VersaTiles: Algebra and Functions*. Match each equation to the solution. Flip over the box to see if you did it correctly.

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Grades 9–12

Data and Interpreting Graphs

Objectives

The student will be able to:

- find the mean, median, and mode of a set of data.
- use data from a table to solve problems.
- use data from a graph to solve problems.

Materials

- *VersaTiles: Statistics and Probability* book (VT8SP)
- VersaTiles

Background Information

Example 1:

The list below shows the height of the basketball players in inches.

75, 69, 68, 69, 74, 70, 72, 74, 68, 72, 70, 68

There are many ways to evaluate a set of data. The **mean** of a set of data is the mathematical average. To find the mean, add all of the data values and divide by the number of data values.

There are 12 data values for this set. The sum of all 12 data values is 849. Therefore, the mean is $\frac{849}{12}$ or approximately 70.75 inches. This means the average height of a player on the team is 70.75 inches.

The **median** of a data set is another way to evaluate data. The median is the middle value when the data values are arranged in order from least to greatest. To find the median, first list the data values in order.

68, 68, 68, 69, 69, 70, 70, 72, 72, 74, 74, 75

Since there are 12 values, there is no true middle value so the median will be the average of the sixth and seventh data values. Both of these values are 70, so the median is 70 inches. This means that there are an equal number of players who are 70 inches or shorter and 70 inches or taller.

Finally, the **mode** of a data set is the value that occurs most often. The value 68 occurs more often than any of the other values, so it is the mode.

68, 68, 68, 69, 69, 70, 70, 72, 72, 74, 74, 75

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Data and Interpreting Graphs

Example 2:

Marco tracked his spending last summer. The table below shows what he did with his earnings.

Spent / Saved	Amount of Money
Savings for College	\$1,120
Spent on Bills/Entertainment	\$288
Other	\$192

What percentage of Marco's earnings went to saving for college?

To determine the percentage of his earnings that went to college savings, first find the total amount of money he earned.

$$\$1,120 + \$288 + \$192 = \$1,600$$

To find the percentage, divide the amount of money for college savings by the total amount of earnings.

$$\frac{\$1,120}{\$1,600} = 0.70$$

Marco saved 70% of his summer earnings for college.

If Marco distributes his money the same, and he plans on earning \$2,400 this summer, how much money will he save for college?

Working backwards, find 70% of \$2,400.

$$0.70 \cdot \$2,400 = \$1,680$$

This summer Marco should be able to save \$1,680 for college.

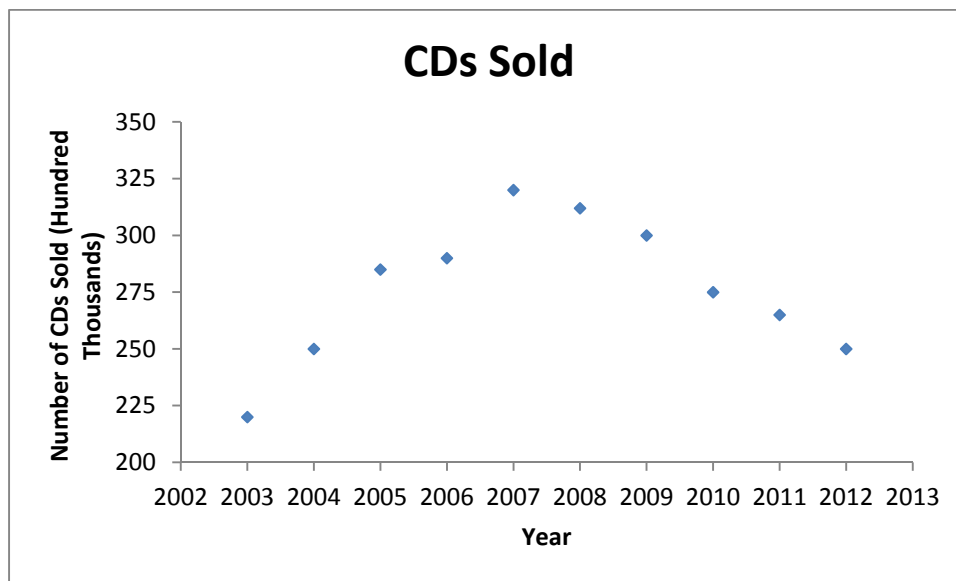
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Grades 9–12

Data and Interpreting Graphs

Example 3:

A local music store has tracked their sales from CDs over the past 10 years. The results are shown in the graph below.



What conclusions can you draw from this graph? What predictions can be made?

Conclusions: Sales of CDs increased from 2003 to 2007. After 2007, the sales of CDs began to decline.

Predictions: Given the data since 2007, it appears that sales are on the decrease. One reason for this could be the digital downloads available from many sources.

Activity

Go to page 1 in *VersaTiles: Statistics and Probability*. Match the problem to the answer. Flip over the box to see if you did it correctly.

Go to page 3 in *VersaTiles: Statistics and Probability*. Match the problem to the answer. Flip over the box to see if you did it correctly.

Go to page 5 in *VersaTiles: Statistics and Probability*. Match the problem to the answer. Flip over the box to see if you did it correctly.

Go to page 12–13 in *VersaTiles: Statistics and Probability*. Match the problem to the answer. Flip over the box to see if you did it correctly.

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Grades 9–12

VersaTiles: Statistics and Probability

Objectives

The student will be able to:

- apply the Fundamental Counting Principle.
- find the number of permutations or combinations of a set of objects.
- find the probability of independent events.

Materials

- *VersaTiles: Statistics and Probability* book (VT8SP)
- VersaTiles

Background Information

Example 1:

There are four appetizers, 10 entrées, and three desserts on a menu. How many combinations of one appetizer, one entrée, and one dessert are possible?

The **Fundamental Principle of Counting** says that if one event can be done in m_1 ways, a second event in m_2 ways, a third event in m_3 , and so on, then the number of ways events one, two, and three can be done is m_1 , m_2 , m_3 .

Applying this to the problem, one appetizer can be chosen in four ways, one entrée can be chosen in 10 ways, and one dessert can be chosen in three ways.

$$4 \cdot 10 \cdot 3 = 120$$

There are 120 ways to choose one appetizer, one entrée, and one dessert.

Example 2:

How many different ways can the first, second, and third place be awarded in a swim meet with eight swimmers?

A **permutation** is all the possible ways of arranging or ordering a set of objects. In this problem, there are eight possible swimmers trying to place in first, second, and third place.

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VersaTiles: Statistics and Probability

Beginning with first place, there are eight different swimmers who can place first. Once first has placed, there are seven swimmers left for second. Finally, there are 6 swimmers left to place in third.

$$\underline{\quad 8 \quad}, \underline{\quad 7 \quad}, \underline{\quad 6 \quad}$$

First Second Third

Therefore, there are $8 \cdot 7 \cdot 6$ or 336 ways that first, second, and third place can be awarded.

Example 3:

Your teacher is choosing four students from the class of 20 for a presentation. How many combinations of four students are possible?

This problem is different in that order does not matter. A **combination** is all possible ways of choosing a group from a set of objects.

Begin by finding the number of permutations from this group. If there are 20 students and four are being chosen, the number of permutations of four from a set of 20 is:

$$20 \cdot 19 \cdot 18 \cdot 17 = 116,280$$

However, the order in which the four are chosen does not matter because it would be the same group of four. To account for this, the number of permutations must be divided by the number of ways a group of four can be ordered.

$$4 \cdot 3 \cdot 2 \cdot 1 = 24$$

Therefore, the number of ways a group of four students can be chosen from a group of 20 is $\frac{116,280}{24} = 4,845$ ways.

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VersaTiles: Statistics and Probability

Example 4:

Find the probability of rolling a six-sided cube and getting a number greater than four and of also flipping heads on a fair coin.

This is an example of an **independent event**. Two events are independent events if one event does not affect the outcome of the other. Rolling a cube and flipping a coin do not affect each other. The probability of two independent events is the product of the probability of each individual event.

Find the probability of the event of rolling a number greater than four. Since there are two numbers on a cube greater than four and a total of six possible outcomes, the probability is $\frac{2}{6}$ or $\frac{1}{3}$.

Next the probability of flipping heads on a fair coin is $\frac{1}{2}$.

Therefore, the probability of rolling a number greater than four and flipping heads on a fair coin is $\frac{1}{3} \cdot \frac{1}{2}$ or $\frac{1}{6}$.

Activity

Go to page 22 in *VersaTiles: Statistics and Probability*. Match each problem with the answer. Flip over the box to see if you did it correctly.

Go to page 23 in *VersaTiles: Statistics and Probability*. Match each problem with the answer. Flip over the box to see if you did it correctly.

Go to page 24 in *VersaTiles: Statistics and Probability*. Match each problem with the answer. Flip over the box to see if you did it correctly.

Go to page 26 in *VersaTiles: Statistics and Probability*. Match each problem with the answer. Flip over the box to see if you did it correctly.

Go to page 27 in *VersaTiles: Statistics and Probability*. Match each problem with the answer. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™

Grades 9–12

Algebra Concept Practice

Objectives

The student will be able to:

- use mental math to multiply.
- perform operations with integers.
- solve expressions and equations involving variables.

Materials

- Learning Wrap-Ups for Pre-Algebra

Background Information

Example 1:

Multiply 6×38 using mental math.

Mental math means simplifying a problem without the use of a calculator. A problem can often be rewritten using numbers that are easier to work with. To find a product of two numbers using mental math, rewrite the multiplication problem as the sum of the products of easier numbers. For example:

$$6 \times 38 = (6 \times 30) + (6 \times 8)$$

$$6 \times 38 = 180 + 48$$

$$6 \times 38 = 228$$

The number 38 was rewritten as $30 + 8$ since it is easier to multiply 6 by 30 and 8 rather than 38.

The problem could also have been written using subtraction.

$$6 \times 38 = (6 \times 40) - (6 \times 2)$$

$$6 \times 38 = 240 - 12$$

$$6 \times 38 = 228$$

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Grades 9–12
Algebra Concept Practice

Example 2:

Simplify: $(-8 + 11)(-2)$

To simplify an expression involving multiple operations, remember to follow the rules for **order of operations**:

1. Simplify within parentheses.
2. Simplify exponents.
3. Perform all multiplication or division from left to right.
4. Perform all addition or subtraction from left to right.

Begin this example by adding -8 and 11 inside the parentheses, then multiply.

$$\begin{aligned}(-8 + 11)(-2) &= (3)(-2) \\ &= -6\end{aligned}$$

Example 3:

Evaluate $-8 \div (x + 5)$ when $x = -3$.

Substitution means to replace a variable with a number or expression equivalent to it. To evaluate the expression for the given value of x , begin by substituting -3 for x .

$$-8 \div (-3 + 5)$$

Simplify inside the parentheses.

$$-8 \div (2)$$

Then divide.

$$-8 \div (2) = -4$$

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Grades 9–12
Algebra Concept Practice

Example 4:

Find y if $3x - 2y = 15$ and $x = -3$.

Begin by substituting -3 for x in the equation.

$$3(-3) - 2y = 15$$

Next simplify the equation.

$$-9 - 2y = 15$$

In order to isolate y , first add nine to both sides of the equation.

$$\begin{array}{rcl} -9 + 9 - 2y & = & 15 + 9 \\ -2y & = & 24 \end{array}$$

Since y is being multiplied by -2 , divide both sides of the equation by -2 to isolate y .

$$\begin{array}{rcl} \frac{-2y}{-2} & = & \frac{24}{-2} \\ y & = & -12 \end{array}$$

Activity

To use Learning Wrap-Ups for Pre-Algebra, hold the Wrap-Ups at the top of the key and select the specific board you want to wrap. Hold the string next to the first problem on the left and wrap it over the board to the correct answer on the right. Wrap the string behind the board to the next problem on the left. Continue the process until you have solved all of the problems. When you have finished, secure the string in the latch at the bottom of the key and see if the string matches the lines on the back. If it does, then you have answered the problems correctly.

The parts of this activity can be completed in any order.

Part 1

Take the Mental Math (yellow keys) Wrap-Up. Complete the first, second, and third keys.

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Grades 9–12
Algebra Concept Practice

Part 2

Take the Addition and Subtraction of Positive and Negative Numbers (green keys). Complete keys 8, 9, and 10. Take the Multiplication and Division of Positive and Negative Numbers (orange keys) Wrap-Ups. Complete keys 8, 9, and 10. For these keys, the variable always represents the sum, difference, product, or quotient.

Part 3

Take the Solving for the Unknown (yellow with red lettering keys). Complete keys 5 and 10. Take the Understanding Algebraic Expressions (blue keys) Wrap-Ups. Complete keys 8, 9, and 10.

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Grades 9–12

Absolute Value

Objectives

The student will be able to:

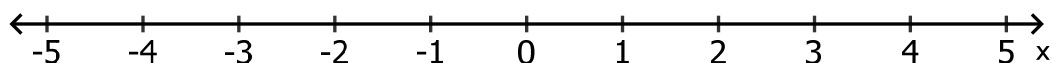
- evaluate the absolute value of a number.
- solve absolute value equations.

Materials

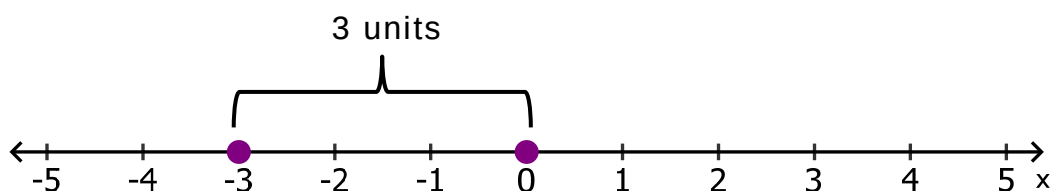
- pair of dice or number cubes

Background Information

The **absolute value** of a number is the distance a number is from zero on the number line.



The symbol for absolute value is two bars. For example, to find $|-3|$ means to find how many units -3 is from zero on the number line.



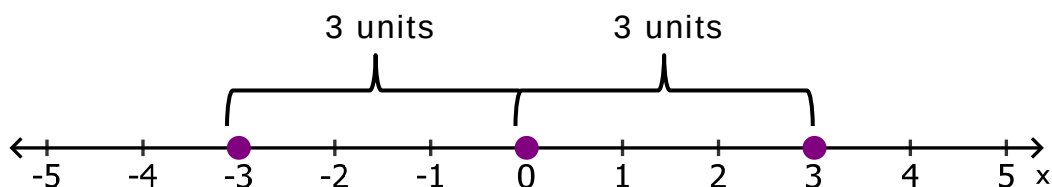
On the number line -3 is plotted as well as zero. The distance from -3 to zero is three units. Therefore, $|-3| = 3$.

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Absolute Value

Is there another number that lies three units away from zero on the number line?



Yes, if you move three units to the right of zero, you are at +3 which is also three units away from zero.

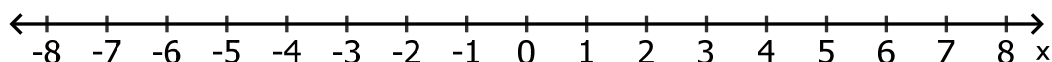
$$|-3| = |3| = 3$$

Since absolute value measures a distance, it is always positive.

Example:

Solve: $|x| = 6$

To solve the equation, look at a number line to help visualize the problem.



Which number or numbers are six units away from zero on the number line?



The solutions to the equation are $x = -6$ and $x = 6$.

Activity

Complete the activities that follow on a separate sheet of paper.

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Grades 9–12

Absolute Value

Part 1

1. Make a table and label the first row as shown below.

a	b	$ a $	$ b $	$ a + b $	$ a + b $
-----	-----	-------	-------	-----------	-------------

2. Roll a pair of dice. Write one of the numbers in the a column and one of the numbers in the b column. Complete the row for the set of values for a and b .
3. Repeat this process four more times. What do you notice about the last two columns for each set of numbers?

Part 2

1. Make a new table and label the first row as shown below.

a	b	$ a $	$ b $	$ a + b $	$ a + b $
-----	-----	-------	-------	-----------	-------------

2. Roll a pair of dice. Write one of the numbers in the a column and one of the numbers in the b column. Now make each a -value negative. Then complete the row for the set of values for a and b .
3. Repeat this process four more times. What do you notice about the last two columns for each set of numbers?

Part 3

1. Make a third table and label the first row as shown below.

a	b	$ a $	$ b $	$ a + b $	$ a + b $
-----	-----	-------	-------	-----------	-------------

2. Roll a pair of dice. Write one of the numbers in the a column and one of the numbers in the b column. Now make both the a - and b -values negative. Then complete the row for the set of values for a and b .
3. Repeat this process four more times. What do you notice about the last two columns for each set of numbers?

Mathematics for SylvanSync™

Grades 9–12

Matrices

Objectives

The student will be able to:

- create a matrix to represent given data.
- calculate the sum or difference of two matrices.
- calculate the product of a matrix and a scalar.

Materials

- none

Background Information

Example 1:

One serving of orange juice has 112 calories, 0.5 grams of fat, 25.79 grams of carbohydrates, and 1.74 grams of protein. One serving of apple juice has 87 calories, 0.2 grams of fat, 21.72 grams of carbohydrates, and 0.11 grams of protein. Use a matrix to represent the data.

A matrix is a rectangular arrangement of numbers enclosed by brackets. To represent the data in a matrix, organize the data in a table.

	Calories	Grams of Fat	Grams of Carbohydrates	Grams of Protein
Serving of Orange Juice	112	0.5	25.79	1.74
Serving of Apple Juice	87	0.2	21.72	0.11

Each row of the matrix will represent a different type of juice, and each column will represent the nutritional facts. The matrix will look like this:

$$A = \begin{bmatrix} 112 & 0.5 & 25.79 & 1.74 \\ 87 & 0.2 & 21.72 & 0.11 \end{bmatrix}$$

A matrix is named by a capital letter, in this case A. Each number in the matrix is called an element of the matrix. The dimensions of a matrix are given by the number of rows by the number of columns. The matrix A has dimensions 2 x 4, said 2 by 4.

Find 2A.

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Grades 9–12

Matrices

When a matrix is being multiplied by a number or a scalar, each element in the matrix must be multiplied by the scalar.

$$2A = \begin{bmatrix} 2 \cdot 112 & 2 \cdot 0.5 & 2 \cdot 25.79 & 2 \cdot 1.74 \\ 2 \cdot 87 & 2 \cdot 0.2 & 2 \cdot 21.72 & 2 \cdot 0.11 \end{bmatrix} = \begin{bmatrix} 224 & 1 & 51.58 & 3.48 \\ 174 & 0.4 & 43.44 & 0.22 \end{bmatrix}$$

What is the meaning of the new matrix?

The resulting matrix represents the number of calories, fat grams, grams of carbohydrates, and grams of protein in two servings of orange juice and apple juice.

Example 2:

One serving of a bagel with cream cheese has 330 calories, 8 grams of fat, 53 grams of carbohydrates, and 11 grams of protein. One serving of scrambled eggs has 365 calories, 26.86 grams of fat, 4.84 grams of carbohydrates, and 24.4 grams of protein. Use a matrix to represent the data.

	Calories	Grams of Fat	Grams of Carbohydrates	Grams of Protein
Serving of Bagel with Cream Cheese	330	8	53	11
Serving of Scrambled Eggs	365	26.86	4.84	24.4

This matrix can be set up similar to the one in example 1.

$$B = \begin{bmatrix} 330 & 8 & 53 & 11 \\ 365 & 26.86 & 4.84 & 24.4 \end{bmatrix}$$

Find $A + B$.

To add or subtract matrices, the dimensions must be the same. Since A and B both have dimensions 2×4 , they can be added. To perform the addition, add each element in matrix A with the corresponding element in matrix B .

$$A + B = \begin{bmatrix} 112 + 330 & 0.5 + 8 & 25.79 + 53 & 1.74 + 11 \\ 87 + 365 & 0.2 + 26.86 & 21.72 + 4.84 & 0.11 + 24.4 \end{bmatrix}$$

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Matrices

The result is:

$$A + B = \begin{bmatrix} 442 & 8.5 & 78.79 & 12.74 \\ 452 & 27.06 & 26.56 & 24.51 \end{bmatrix}$$

What is the meaning of this matrix?

The first row represents the nutritional information for one serving of orange juice and one serving of a bagel with cream cheese. The second row is the nutritional information for one serving of apple juice and one serving of scrambled eggs.

Example 3:

$$\text{If } C = \begin{bmatrix} -9 & \frac{8}{9} \\ 0 & 12.2 \\ -4 & 6 \end{bmatrix} \text{ and } D = \begin{bmatrix} -5 & \frac{1}{3} \\ -7.4 & 5 \\ 1 & -2 \end{bmatrix}, \text{ find } C - D.$$

The matrix D can be subtracted from matrix C because they have the same dimensions. Subtract each element in D from the corresponding element in C . Watch the signs and remember the rules for subtracting fractions.

$$C - D = \begin{bmatrix} -9 - (-5) & \frac{8}{9} - \frac{1}{3} \\ 0 - (-7.4) & 12.2 - 5 \\ -4 - 1 & 6 - (-2) \end{bmatrix} = \begin{bmatrix} -9 + 5 & \frac{8}{9} - \frac{1}{3} \\ 0 + 7.4 & 12.2 - 5 \\ -4 - 1 & 6 + 2 \end{bmatrix} = \begin{bmatrix} -4 & \frac{5}{9} \\ 7.4 & 7.2 \\ -5 & 8 \end{bmatrix}$$

Mathematics for SylvanSync™

Grades 9–12

Matrices

Activity

Complete the activities that follow on a separate sheet of paper.

Part 1

A sports car gets 19 miles per gallon for city driving and 30 miles per gallon for highway driving. A hybrid sedan gets 47 miles per gallon for city driving and 47 miles per gallon for highway driving. A truck gets 21 miles per gallon for city driving and 25 miles per gallon for highway driving.

1. Create a matrix with dimensions 2×3 to represent the data given. Call it matrix A .
2. What does each row represent in the matrix?
3. What does each column represent in the matrix?
4. Find $1.10A$.
5. What is the meaning of the matrix found in question 4?

Part 2

$$M = \begin{bmatrix} -3 & 0.75 \\ 9 & 12 \\ \frac{2}{5} & -5 \\ 0 & 1 \end{bmatrix} \quad N = \begin{bmatrix} 2 & -7 \\ 11 & -\frac{1}{2} \\ -\frac{3}{10} & 8 \\ -6 & 5.2 \end{bmatrix}$$

1. In order to add or subtract matrices, what must be true?
2. Find $M + N$.
3. Find $N + M$.
4. Are your answers to question 2 and question 3 different? What property does this represent?
5. Is $M - N = N - M$? If not, find $M - N$ and $N - M$.

Mathematics for SylvanSync™

Grades 9–12

Equation of a Line

Objectives

The student will be able to:

- write an equation of a line in slope-intercept form.
- write an equation of a line in standard form.
- express a given equation in either form.

Materials

- none

Background Information

Example 1:

To rent a snow-cone machine, there is a base fee of \$50. In addition to the base fee, there is an hourly fee of \$12 per hour rented. Write an equation to represent the cost, y , of renting the snow-cone machine for x hours.

The cost function is linear because the cost has a constant rate of increase of \$12. The equation is $y = 12x + 50$.

This equation is written in **slope-intercept form**, $y = mx + b$ where m is the slope and b is the y -intercept.

Example 2:

Write the equation $y = 12x + 50$ in standard form.

Standard form is $Ax + By = C$ where A , B , and C are integers and $A > 0$. To rewrite the equation in standard form, move the x term to the same side as the y term by subtracting $12x$ from both sides.

$$\begin{aligned} y &= 12x + 50 \\ -12x + y &= 12x - 12x + 50 \\ -12x + y &= 50 \end{aligned}$$

Multiply both sides of the equation by -1 so that the coefficient of x is positive.

$$\begin{aligned} -1(-12x + y) &= (50)(-1) \\ 12x - y &= -50 \end{aligned}$$

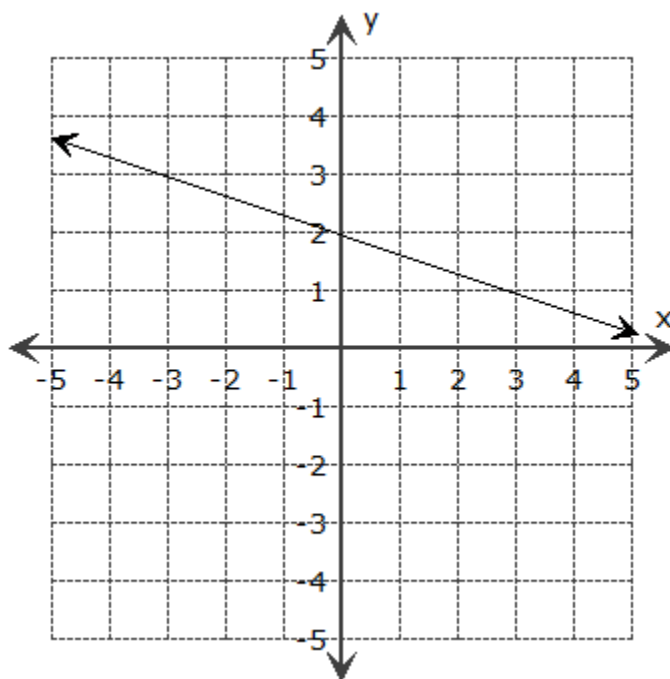
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Grades 9–12

Equation of a Line

Example 3:

Write the equation of the line graphed below in both slope-intercept form and standard form.



To write the equation of a line in slope-intercept form, it is necessary to know the slope and the y-intercept.

The points (0, 2) and (3, 1) are two points on the line. **Slope** measures the steepness of a line. Use these two points to find the slope of the line. Recall the formula for finding slope.

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

Let $x_1 = 0$, $y_1 = 2$, $x_2 = 3$, and $y_2 = 1$. Substitute these values in the slope formula.

$$\text{slope} = \frac{1-2}{3-0} = \frac{-1}{3} = -\frac{1}{3}$$

Now find the **y-intercept** of the graph. The y-intercept is the y-coordinate of the point where the line intersects the y-axis. Looking at the graph, the y-intercept is 2.

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Equation of a Line

The slope of the line is $-\frac{1}{3}$ and the y -intercept is 2. Slope-intercept form is $y = mx + b$. Substituting $-\frac{1}{3}$ for m and 2 for b , the equation is $y = -\frac{1}{3}x + 2$.

To write this equation in standard form, begin by multiplying both sides of the equation by three to eliminate the fraction.

$$(3)(y) = \left(-\frac{1}{3}x + 2\right)3$$

$$3y = -x + 6$$

Next add x to both sides of the equation.

$$x + 3y = -x + x + 6$$

$$x + 3y = 6$$

The equation of the line graphed in slope-intercept form is $y = -\frac{1}{3}x + 2$ and $x + 3y = 6$ in standard form.

Example 4:

Write the equation $9x - 9x - 5y = 15$ in slope-intercept form.

To write the equation in slope-intercept form, y must be isolated. Begin by subtracting $9x$ from both sides of the equation.

$$9x - 9x - 5y = 15 - 9x$$

$$-5y = -9x + 15$$

To isolate y , divide both sides of the equation by -5 . On the left side, each term is divided by -5 .

$$\frac{-5y}{-5} = \frac{-9x + 15}{-5}$$

$$y = \frac{-9x}{-5} + \frac{15}{-5}$$

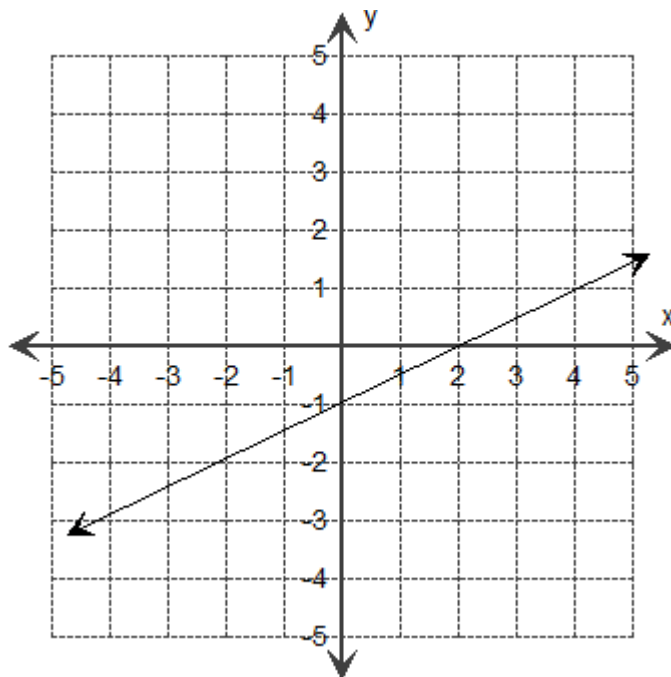
$$y = \frac{9}{5}x - 3$$

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Grades 9–12
Equation of a Line

Activity

Complete the activities that follow on a separate sheet of paper.

Part 1



1. Write the equation of the line graphed in slope-intercept form.
2. Using your answer to question 1, write the equation in standard form.
3. Write the equation of the line through the points $(0, 11)$ and $(-10, 8)$ in slope-intercept form.
4. Using your answer to question 3, write the equation in standard form.

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Equation of a Line

Part 2

For each equation given, if the equation is in slope-intercept form, write it in standard form. If the equation is in standard form, write it in slope-intercept form.

1. $y = \frac{-3}{4}x - 7$

2. $5x + 6y = 20$

3. $8y = 16$

4. $y = 3x - 2$

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Grades 9–12

Polynomials

Objectives

The student will be able to:

- classify polynomials.
- determine the degree of a polynomial.
- simplify polynomials.

Materials

- none

Background Information

A polynomial is an expression that is one term or the sum of terms in the form ax^n , where n is a nonnegative integer.

Take a look at the table below to view some examples of polynomials and some non-examples.

Polynomials	Not Polynomials
12	$\frac{9}{x}$
$9x^2 - x + 6$	$5\sqrt{x} + 4x^2$
$-15x^4y^2 + 8xy - 9$	$10x^{-4} + x^3 + 7$

A **term** of a polynomial is an expression which stands alone or is separated from another term by addition or subtraction signs. The **degree of a term** is the sum of the exponents of each variable in the term.

Example 1:

The polynomial 12 is a term. The degree of the term 12 is 0 since $12 = 12x^0$.

Example 2:

The polynomial $-15x^4y^2 + 8xy - 9$ has three terms. They are $-15x^4y^2$, $8xy$, and -9 . The degree of the first term, $-15x^4y^2$, is $4 + 2$ or 6 since the exponent of x is 4 and the exponent of y is 2. The degree of the second term, $8xy$, is 2. The degree of the last term, -9 , is 0.

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Grades 9–12

Polynomials

The **degree of a polynomial** is the degree of the term with the highest degree. The polynomial $-15x^4y^2 + 8xy - 9$ had terms with degrees of 4, 2, and 0. Therefore, the degree of the polynomial is 4.

Polynomials can be classified by the degree of the polynomial and by the number of terms in the polynomial.

The table below shows how to classify a polynomial by its degree.

Name	Degree	Examples
Constant	0	12
Linear	1	$-8x + 5$
Quadratic	2	$9x^2 - x + 6$
Cubic	3	$x^3 + 2x^2 - 8x + 2$
Quartic	4	$5xy^3 + 2xy - 11x + 4$

This table shows how to classify a polynomial by the number of terms.

Name	Number of Terms	Examples
Monomial	1	12
Binomial	2	$-8x + 5$
Trinomial	3	$9x^2 - x + 6$

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Polynomials

Example 3:

Classify the polynomial $-8xy^2z + 9xyz - 11$ by the degree of the polynomial and the number of terms.

The degree of the polynomial is 4 since the term $-8xy^2z$ has a degree of 4, and it has 3 terms. It is a quartic trinomial.

Example 4:

Add: $(9x^2 + 6x - 7) + (x^2 - 8x - 4)$

To add two polynomials, combine like terms. **Like terms** are terms that have the same variable part including exponents. Rewrite the sum by grouping the like terms.

$$(-9x^2 + x^2) + (6x + (-8x)) + (-7 + (-4))$$

Simplify by combining the coefficients.

$$(-8x^2) + (-2x) + (-11) = -8x^2 - 2x - 11$$

Example 5:

Subtract: $(3y^4 - 6y^2 + 8) - (9y^4 - 5y^2 + y - 3)$

Recall that subtraction can be rewritten as addition of the opposite. So the problem can be rewritten like this:

$$(3y^4 - 6y^2 + 8) + (9y^4 - 5y^2 + y - 3)$$

Notice every term in the second polynomial is now the opposite. Now group like terms and simplify.

$$(3y^4 + (-9y^4)) + (-6y^2 + 5y^2) + (-y) + (8 + 3) = -6y^4 - y^2 - y + 11$$

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Grades 9–12

Polynomials

Activity

Complete the activities that follow on a separate sheet of paper.

Part 1

Classify each polynomial by the degree and the number of terms.

1. $3xy^3 + 5xyz$

2. -5

3. $-13 + 5a^2 - 10a^3$

4. $12m^2 - 8m$

Part 2

Simplify.

1. $(-6x^2 - 8) + (10x^2 - 3x - 5)$

2. $(3x^2 - 5xy - 7y^2) - (-x^2 + 3xy + 10y^2)$

3. $(a^3 - 3a^2 + 4a - 11) + (6a^3 - a^2 + a + 5)$

4. $(5n^3 + 6n^2 - 9) - (9n^3 - 4n^2 + 5n - 1)$

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Grades 9–12

Multiply Polynomials

Objectives

The student will be able to:

- find the product of a trinomial and a monomial.
- find the product of a trinomial and a binomial.
- find the product of two trinomials.

Materials

- none

Background Information

Example 1:

Multiply: $9x(-6x^2 + 7x - 2)$

To multiply, distribute $9x$ to each term of the polynomial.

$$(9x)(-6x^2) + (9x)(7x) + (9x)(-2)$$

$$\text{Simplify: } -54x^3 + 63x^2 - 18x$$

Recall that when multiplying expressions with the same variable, the exponents are added.

Example 2:

Multiply: $(5a - 2)(3a^5 + a^3 - 2a)$

To multiply a binomial by a trinomial, distribute each term of the binomial to the trinomial.

$$5a(3a^5 + a^3 - 2a) - 2(3a^5 + a^3 - 2a)$$

Then apply the distributive property again.

$$(5a)(3a^5) + (5a)(a^3) + (5a)(-2a) + (-2)(3a^5) + (-2)(a^3) + (-2)(-2a)$$

$$15a^6 + 5a^4 - 10a^2 - 6a^5 - 2a^3 + 4a$$

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Grades 9–12

Multiply Polynomials

Rewrite the product in descending order:

$$15a^6 - 6a^5 + 5a^4 - 2a^3 - 10a^2 + 4a$$

Example 3:

Multiply: $(3x^2 - 5x + 6)(-x^2 + 2x - 3)$

Distribute each term in the first trinomial to the second trinomial.

$$3x^2(-x^2 + 2x - 3) - 5x(-x^2 + 2x - 3) + 6(-x^2 + 2x - 3)$$

Use the distributive property again and simplify.

$$(3x^2)(-x^2) + (3x^2)(2x) + (3x^2)(-3) + (-5x)(-x^2) + (-5x)(2x) + (-5x)(-3) \\ + (6)(-x^2) + (6)(2x) + (6)(-3)$$

$$-3x^4 + 6x^3 - 9x^2 + 5x^3 - 10x^2 + 15x - 6x^2 + 12x - 18$$

Combine like terms.

$$-3x^4 + 11x^3 - 25x^2 + 27x - 18$$

Activity

Complete the activities that follow on a separate sheet of paper.

Multiply:

1. $(4x + 5)(x^2 - x + 3)$
2. $(x + 2y)(3x - 4y + 5)$
3. $-5(6m^4 - 3m^2 + 2)$
4. $6xyz(x - 7y + 2z)$
5. $(x^2 + 3x - 4)(x^2 - 4x + 2)$
6. $(-3a + b - 6c)(a - 5b + 3c)$

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Grades 9–12

Multiplying Binomials

Objectives

The student will be able to:

- determine and apply a pattern to calculate the product of the sum and difference of two terms.
- determine and apply a pattern to calculate the square of a binomial.
- determine and apply a pattern to calculate the product of any two binomials.

Materials

- none

Background Information

Example 1:

Multiply: $(3x - 2y)(3x + 2y)$

To multiply, distribute each term in the first binomial to the second binomial.

$$3x(3x - 2y) - 2y(3x + 2y)$$

Use the distributive property again.

$$(3x)(3x) + (3x)(2y) + (-2y)(3x) + (-2y)(2y)$$

Simplify and combine like terms. Like terms are terms that have the same variable part.

$$9x^2 + 6xy - 6xy - 4y^2 = 9x^2 - 4y^2$$

Example 2:

Multiply: $(4m - n)^2$

To square a binomial means to multiply the binomial by itself.

$$(4m - n)(4m - n)$$

Applying the distributive property, you have:

$$4m(4m - n) - n(4m - n)$$

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Grades 9–12

Multiplying Binomials

Then apply the distributive property again.

$$(4m)(4m) + (4m)(-n) + (-n)(4m) + (-n)(-n)$$

Simplifying and combining like terms:

$$16m^2 - 4mn - 4mn + n^2 = 16m^2 - 8mn + n^2$$

Example 3:

Multiply: $(3x - 4)(x + 5)$

Distribute each term in the first binomial to the second binomial.

$$3x(x + 5) - 4(x + 5)$$

Use the distributive property again and simplify.

$$(3x)(x) + (3x)(5) + (-4)(x) + (-4)(5) = 3x^2 + 15x - 4x - 20 = 3x^2 + 11x - 20$$

Activity

Complete the activities that follow on a separate sheet of paper.

Part 1

Multiply:

1. $(x - 2)(x + 2)$
2. $(x + 6)(x - 6)$
3. $(2x - 3)(2x + 3)$
4. $(x - 7y)(x + 7y)$
5. $(y + 5)(y - 5)$

Look back at your work for questions 1 through 5. Answer the questions.

7. What is similar about each pair of binomials being multiplied? How many terms are in each answer? What pattern do you see with all of the coefficients and constants in the answers?
6. Write a pattern that can be used to multiply $(m - n)(m + n)$.

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Grades 9–12
Multiplying Binomials

Part 2

Multiply:

1. $(x + 5)^2$
2. $(a - 4)^2$
3. $(2x - 3)^2$
4. $(3y + 4)^2$
5. $(5x - 3y)^2$

Look back at your work for questions 1 through 5. Answer the questions.

6. What is similar about each pair of binomials being multiplied? How many terms are in each answer?
7. Write a pattern that can be used to multiply $(m - n)^2$ and $(m + n)^2$.

Part 3

For questions 1 through 4, multiply.

1. $(x + 4)(x - 6)$
2. $(3x + 2)(x + 5)$
3. $(a - 7)(4a + 3)$
4. $(x - 8)(2x - 5)$
5. Write a pattern that can be used to multiply $(a - b)(c - d)$.
6. Write a pattern that can be used to multiply $(a + b)(c + d)$.
7. Write a pattern that can be used to multiply $(a - b)(c + d)$.

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Grades 9–12

Factoring Binomials and Trinomials

Objectives

The student will be able to:

- write a polynomial as the product of its GCF and another factor.
- factor special binomials by recognizing a pattern.
- factor special trinomials by recognizing a pattern.

Materials

- none

Background Information

To **factor** a polynomial means to rewrite it as a product of its factors. When factoring, first look to see if there is a **greatest common factor (GCF)**. The GCF is the greatest factor that the terms in the expression have in common. Take a look at an example.

Example 1:

Factor: $24m^2n - 8m^3n + 4m^2n^2$ $24m^2n - 8m^3n + 4m^2n$

Begin by looking at the numerical part of each term. The numbers 24, 8, and 4 have common factors of 1, 2, and 4. The greatest numerical factor is 4.

Now look at the variables. For the variable m , there is an m^2 , m^3 , and m^2 . The greatest common factor for m is m^2 . For the variable n , there is n , n , and n^2 . The greatest common factor for the n variable is n .

Multiplying these factors together, the greatest common factor is $4m^2n$. Rewrite each term as a product using the GCF.

$$24m^2n - 8m^3n + 4m^2n^2 = (4m^2n)(6) - (4m^2n)(2m) + (4m^2n)(n)$$

Now, factor out the GCF to write it as a product.

$$(4m^2n)(6) - (4m^2n)(2m) + (4m^2n)(n) = (4m^2n)(6 - 2m + n)$$

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Grades 9–12
Factoring Binomials and Trinomials

Example 2:

Factor: $x^2 - 9$

The binomial $x^2 - 9$ is a special binomial. Notice how each term can be rewritten as an expression being squared: $x^2 - 9 = (x)^2 - (3)^2$. This is an example of a difference of two squares. The pattern for factoring a difference of two squares is:

$$a^2 - b^2 = (a - b)(a + b)$$

For the binomial $x^2 - 9$ which was rewritten as $(x)^2 - (3)^2$, $a = x$ and $b = 3$.

$$x^2 - 9 = (x)^2 - (3)^2 = (x - 3)(x + 3)$$

Notice how the factored form is two binomials with the same terms and different signs.

Note the sum of two squares **does not** factor. An example of this is $x^2 + 9$.

Example 3:

Factor: $9a^2 - 30a + 25$

The trinomial $9a^2 - 30a + 25$ has some special characteristics. Notice how the first and last terms have numbers that are perfect squares, 9 and 25. This trinomial can also be factored using a pattern.

$$a^2 - 2ab + b^2 = (a - b)(a - b) = (a - b)^2$$

To help see the pattern, rewrite the trinomial.

$$9a^2 - 30a + 25 = (3a)^2 - 2(3a)(5) + (5)^2$$

Therefore, in factored form the trinomial is $(3a - 5)(3a - 5) = (3a - 5)^2$.

Similarly, the trinomial $a^2 + 2ab + b^2 = (a + b)(a + b) = (a + b)^2$. So,
 $9a^2 - 30a + 25 = (3a - 5)^2$.

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Grades 9–12
Factoring Binomials and Trinomials

Activity

Complete the activities that follow on a separate sheet of paper.

Part 1

Factor completely. If not factorable, write “Not Factorable.”

1. $5x - 15$
2. $yz^3 - 3yz^2 + 2y$
3. $6x^3 - 24x^2 + 6x$
4. $3a^2b^3 + 9ab^2 - 12a^2b^2$

Part 2

Factor completely. If not factorable, write “Not Factorable.”

1. $x^2 - 100$
2. $9a^2 - 49$
3. $16x^2 - y^2$
4. $z^2 + 16$

Part 3

Factor completely. If not factorable, write “Not Factorable.”

1. $x^2 - 6x + 9$
2. $m^2 + 8m + 16$
3. $4y^2 + 20y + 25$
4. $16b^2 - 8b + 1$

Part 4

Factor completely. If not factorable, write “Not Factorable.”

1. $x^3 - 4x$
2. $6a^2 - 54$
3. $2y^2 - 20y + 50$
4. $8a^2 - 8ab + 8b^2$

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Grades 9–12

Polynomial Division

Objectives

The student will be able to:

- divide a polynomial by a monomial.
- use long division to divide a polynomial by a binomial.

Materials

- none

Background Information

Example 1:

Divide: $\frac{6x^2 - 2x + 3}{2x}$

To divide a polynomial by a monomial, divide each term of the dividend by the monomial.

$$\frac{6x^2 - 2x + 3}{2x} = \frac{6x^2}{2x} - \frac{2x}{2x} + \frac{3}{2x}$$

To simplify, recall the rules of exponents when dividing expressions with the same variable. Subtract the exponents to simplify. Also, simplify each fraction.

$$\frac{6x^2}{2x} - \frac{2x}{2x} + \frac{3}{2x} = \frac{3x^{2-1}}{1} - \frac{1x^{1-1}}{1} + \frac{3}{2x} = 3x - 1 + \frac{3}{2x}$$

Just as long division can be used to divide numbers, it can also be used to divide polynomials by polynomials.

Example 2:

Divide: $\frac{2x^3 - 5x^2 - 2x - 3}{x - 3}$

Before performing the long division, take a look at the following definitions. The **dividend** is the quantity that is being divided. The **divisor** is the quantity that divides the dividend. The result when dividing is called the **quotient**. The polynomial $2x^3 - 5x^2 - 2x - 3$ is the dividend and the binomial $x - 3$ is the divisor.

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Grades 9–12

Polynomial Division

The polynomials must be written in standard form and if a term is missing, write a placeholder for that term.

$$x - 3 \overline{) 2x^3 - 5x^2 - 2x - 3}$$

Begin by dividing the first term of the dividend, $2x^3$, by the first term of the divisor x . The result is x^2 . Place it above the first term of the dividend.	$\begin{array}{r} 2x^2 \\ x - 3 \overline{) 2x^3 - 5x^2 - 2x - 3} \end{array}$
Next multiply the divisor by $2x^2$ and write it under the dividend.	$\begin{array}{r} 2x^2 \\ x - 3 \overline{) 2x^3 - 5x^2 - 2x - 3} \\ \underline{2x^3 - 6x^2} \end{array}$
Subtract the product from the dividend. Be sure to distribute the subtraction sign. Bring down all the terms from the dividend.	$\begin{array}{r} 2x^2 \\ x - 3 \overline{) 2x^3 - 5x^2 - 2x - 3} \\ \underline{-(2x^3 - 6x^2)} \\ 0x^3 + x^2 - 2x - 3 \end{array}$
Repeat the process by dividing x^2 by x .	$\begin{array}{r} 2x^2 + x \\ x - 3 \overline{) 2x^3 - 5x^2 - 2x - 3} \\ \underline{-(2x^3 - 6x^2)} \\ 0x^3 + x^2 - 2x - 3 \\ \underline{-(x^2 - 3x)} \\ 0x^2 + x - 3 \end{array}$
Repeat again by dividing x by x .	$\begin{array}{r} 2x^2 + x + 1 \\ x - 3 \overline{) 2x^3 - 5x^2 - 2x - 3} \\ \underline{-(2x^3 - 6x^2)} \\ 0x^3 + x^2 - 2x - 3 \\ \underline{-(x^2 - 3x)} \\ 0x^2 + x - 3 \\ \underline{-(x - 3)} \\ 0x - 0 \end{array}$

The quotient is $2x^2 + x + 1$, and there is no remainder. Therefore, $(x - 3)$ is a factor of $2x^3 - 5x^2 - 2x - 3$.

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Grades 9–12
Polynomial Division

Example 3:

Divide: $(x^3 + 4x^2 + 7x - 9) \div (x + 3)$

$$\begin{array}{r}
 x^2 + x + 4 - \frac{21}{x+3} \\
 x + 3 \overline{) x^3 + 4x^2 + 7x - 9} \\
 \underline{-(x^3 + 3x^2)} \\
 0x^3 + x^2 + 7x - 9 \\
 \underline{-(x^2 + 3x)} \\
 0x^2 + 4x - 9 \\
 \underline{-(4x + 12)} \\
 0x - 21
 \end{array}$$

In this division problem, there was a remainder of -21 . The quotient can be written as $x^2 + x + 4 - \frac{21}{x+3}$.

Example 4:

Divide: $(2x^3 - 9x^2 + 27) \div (x - 2)$

Notice in the dividend there is a term missing. There is no x term. Therefore, when performing the long division, the polynomial $2x^3 - 9x^2 + 27$ will be written as $2x^3 - 9x^2 + 0x + 27$ to account for the missing term.

$$\begin{array}{r}
 2x^2 - 5x - 10 + \frac{7}{x-2} \\
 x - 2 \overline{) 2x^3 - 9x^2 + 0x + 27} \\
 \underline{-(2x^3 - 4x^2)} \\
 0x^3 - 5x^2 + 0x + 27 \\
 \underline{-(-5x^2 + 10x)} \\
 0x^2 - 10x + 27 \\
 \underline{-(10x + 20)} \\
 0x + 7
 \end{array}$$

The quotient is $2x^2 - 5x - 10 + \frac{7}{x-2}$.

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Grades 9–12
Polynomial Division

Activity

Complete the activities that follow on a separate sheet of paper.

Part 1

Divide each polynomial by each monomial.

7. $\frac{x^3 - 4x^2 + 7x}{x}$

8. $(8x^2 + 5x + 3) \div (4x)$

9. $\frac{6x - x^3 + 12x^2 + 4}{2x}$

10. $(6x^4 - 2x^3 + 7x^2) \div (3x^2)$

11. $\frac{7x^3 - 14x^2 + 5x - 1}{7x}$

Part 2

Use long division to perform division.

8. $(x^4 - 7x^3 + 4x^2 + 7x - 9) \div (x - 2)$

9. $\frac{x^2 - 2x + 3}{x - 1}$

10. $(x^3 - 1) \div (x + 1)$

11. $\frac{6x^2 + x - 35}{2x + 5}$

12. $(3x^3 - 7x^2 + 2x + 3) \div (x - 1)$

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Grades 9–12

Pythagorean Theorem Model

Objectives

The student will be able to:

- explore the Pythagorean Theorem using a model.

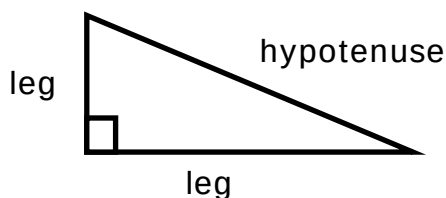
Materials

- Pythagorean Theorem model
- grid paper
- scissors
- protractor
- strips of paper: 2 inches long, 3 inches long, 3 inches long, and 4 inches long

Background Information

The **Pythagorean Theorem** states that the sum of the squares of the legs of a right triangle is equal to the square of the hypotenuse. The **legs** of a right triangle are the sides opposite the acute angles. The **hypotenuse** is the side opposite the right angle and is always the longest side of the triangle.

The theorem can be written as an equation as $leg^2 + leg^2 = hypotenuse^2$ or $a^2 + b^2 = c^2$ where a and b are the legs and c is the hypotenuse.



$$leg^2 + leg^2 = hypotenuse^2$$

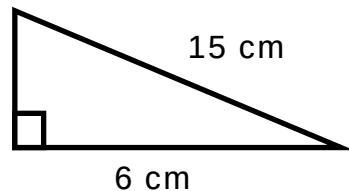
$$a^2 + b^2 = c^2$$

The Pythagorean Theorem is true only for right triangles. Recall **acute triangles** have three acute angles. **Obtuse triangles** have one obtuse angle.

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Pythagorean Theorem Model

Example:

Find the missing side in the triangle below.



The missing side is a leg. Let x represent the length of the leg.

$$x^2 + 6^2 = 15^2$$

To solve the equation, evaluate exponents first. Then isolate x^2 .

$$x^2 + 6^2 = 15^2$$

$$x^2 + 36 = 225$$

$$x^2 + 36 - 36 = 225 - 36$$

$$x^2 = 189$$

To solve for x , take the square root of both sides.

$$x^2 = 189$$

$$\sqrt{x^2} = \sqrt{189}$$

$$x \approx 13.75$$

The length of the missing leg is approximately 13.75 cm.

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Grades 9–12

Pythagorean Theorem Model

Activity

Complete the activities that follow on a separate sheet of paper.

Part 1

Look at the Pythagorean Theorem model. On the grid paper, you will be making a Pythagorean Theorem model similar to this.

1. On the grid paper, draw the legs of a right triangle. Each line should be drawn along a grid line. Draw one leg that is 8 units long, and one leg that is 15 units long. Connect the ends of the legs to draw the hypotenuse.
2. Draw a square on each leg of the triangle. From a different sheet of grid paper, cut a square to place on the hypotenuse of the triangle.
3. Count the number of grid squares in each larger square. Record your answers.
4. Write an equation using the number of smaller squares.
5. Write the equation again, this time using exponential expressions for the numbers of squares.
6. Does the relationship you found in question 5 agree with the Pythagorean Theorem? Explain.

Part 2

1. Make a triangle with the 2 inch, 3 inch, and 4 inch strips of paper. Tape the strips onto a new piece of grid paper.
2. Test if the triangle formed is a right triangle. In the equation $a^2 + b^2 = c^2$, let $a = 2$, $b = 3$, and $c = 4$. Simplify each side of the equation. What is the result? Is it a true statement? Explain.
3. Since the equation $a^2 + b^2 = c^2$ is not true for the triangle, you can conclude that it is NOT a right triangle. Use the protractor to determine if it is an acute or obtuse triangle.

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Grades 9–12
Pythagorean Theorem Model

Part 3

1. Make a triangle with the 3 inch, 3 inch, and 4 inch strips of paper. Tape the strips onto a new piece of grid paper.
2. Test if the triangle formed is a right triangle. In the equation $a^2 + b^2 = c^2$, let $a = 3$, $b = 3$, and $c = 4$. Simplify each side of the equation. What is the result? Is it a true statement?
3. Since the equation $a^2 + b^2 = c^2$ is not true for the triangle, you can conclude that it is NOT a right triangle. Use the protractor to determine if it is an acute or obtuse triangle.

Part 4

Your conclusions from Part 2 and Part 3 can be generalized.

1. Look back at Part 2. If $a^2 + b^2 < c^2$ in a triangle with the largest side being c , what kind of triangle is it? Create a different triangle and test if this statement still holds true.
2. Look back at Part 3. If $a^2 + b^2 > c^2$ in a triangle with the largest side being c , what kind of triangle is it? Create a different triangle and test if this statement holds true.

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Similar Polygons

Objectives

The student will be able to:

- identify corresponding parts of similar polygons.
- create and name similar polygons.

Materials

- grid paper

Background Information

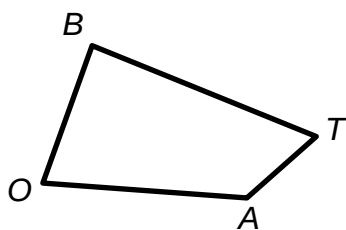
Two polygons are **similar** if corresponding angles are congruent and the lengths of corresponding sides are proportional. Similar polygons are named by listing corresponding vertices in the same order and using the similarity symbol $\sim$.

Example 1:

Use the similarity statement to draw the figures.

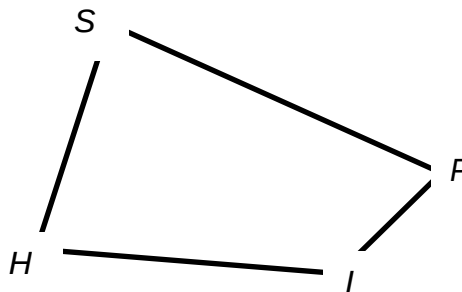
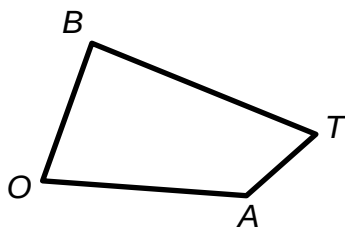
Quadrilateral $BOAT \sim$ Quadrilateral $SHIP$

Draw a four-sided polygon labeled with vertices B , O , A , and T . Be sure to move clockwise or counterclockwise as you label the vertices.



Now draw another quadrilateral with the same shape, but with a different size.

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Similar Polygons



To label the vertices of the second quadrilateral, match the corresponding parts. Notice in the similarity statement below, corresponding vertices are shown in the same color.

Quadrilateral **BOAT** ~ Quadrilateral **SHIP**

Since the polygons are similar, the following statements are true:

$$\angle B \cong \angle S$$

$$\angle O \cong \angle H \quad \text{and} \quad \frac{BO}{SH} = \frac{OA}{HI} = \frac{AT}{IP} = \frac{BT}{SP}$$

$$\angle A \cong \angle I$$

$$\angle T \cong \angle P$$

Example 2:

Given: Quadrilateral $BOAT \sim$ Quadrilateral $SHIP$, $BO = 4$, $OA = 9$, and $SH = 10$. What is HI ?

Using the fact that corresponding sides are proportional, the equation

$\frac{BO}{SH} = \frac{OA}{HI}$ can be used to find HI . Substituting the known values, you have:

$$\frac{4}{10} = \frac{9}{HI}$$

To solve the proportion, multiply the cross products.

$$4 \cdot HI = 10 \cdot 9$$

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Similar Polygons

Simplify and solve for HI :

$$4 \cdot HI = 90$$

$$\frac{4 \cdot HI}{4} = \frac{90}{4}$$

$$HI = 22.5$$

Activity

Complete the activities that follow on a separate sheet of grid paper.

Part 1

1. Draw quadrilateral HINT similar to quadrilateral CLUE.
2. Label the figures with possible angle measures to show they are similar. (Recall that the sum of the angles in a quadrilateral is 360 degrees.)
3. Label the lengths of the sides to show they are similar.
4. What is the ratio of side lengths HI to CL ? Is it the same as NT to UE ?

Part 2

1. Draw two similar quadrilaterals. Name the quadrilaterals such that they are synonyms. If necessary, consult a dictionary or thesaurus. Write the similarity statement for the figures.
2. Label the lengths of the sides of the quadrilaterals such that the ratio of sides is 2 to 3.

Part 3

1. Draw two similar triangles. Name the triangles such that they are synonyms. If necessary, consult a dictionary or thesaurus. Write the similarity statement for the figures.
2. Label the lengths of the sides of the triangles.
3. What is the ratio of the lengths of the sides?